

VILHELMINA, Sweden

WMO: 022450

Lat: 64.580N

Lon: 16.839E

Elev: 349

StdP: 97.20

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-28.8	-35.9	0.1	-31.9	-32.3	0.2	-28.8	9.7	0.7	8.2	-0.2	0.3	110	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	24.7	15.7	22.6	14.9	20.7	14.1	17.1	22.1	16.1	20.6	15.0	19.3	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.4	18.7	14.2	10.5	17.4	13.1	9.8	16.6	49.4	22.2	46.1	20.7	43.1	19.2	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.0	-35.2	26.5	2.6	2.2	-37.1	28.0	-38.6	29.3	-40.1	30.5	-42.0	32.1		
(5)			-35.3	18.3	2.6	1.3	-37.2	19.2	-38.7	20.0	-40.2	20.7	-42.0	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-11.2	-10.2	-6.3	0.2	6.0	10.9	14.0	11.8	7.4	1.1	-4.8	-9.3		
	DBStd	10.30	8.23	7.75	6.15	4.00	3.74	3.49	2.99	3.07	3.27	4.23	6.22	8.10		
	HDD10.0	3612	658	564	505	295	133	29	2	16	89	277	444	599		
	HDD18.3	6383	916	798	763	545	381	224	137	205	327	535	694	857		
	CDD10.0	277	0	0	0	0	11	56	127	71	12	1	0	0		
	CDD18.3	6	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH23.3	116	0	0	0	0	3	26	70	17	0	0	0	0		
	CDH26.7	11	0	0	0	0	0	1	9	1	0	0	0	0		
(14) Wind	WSAvg	2.4	2.0	2.1	2.6	2.6	2.8	2.9	2.4	2.2	2.5	2.3	1.9	1.9		
(15) Precipitation	PrecAvg	610	39	30	29	33	46	59	94	77	60	51	49	40		
	PrecMax	859	61	55	50	51	89	97	168	171	140	124	100	69		
	PrecMin	435	7	8	9	13	16	9	25	28	15	10	21	8		
	PrecStd	106	14	11	12	13	21	23	34	43	28	28	22	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	5.1	7.2	16.1	23.4	26.2	28.1	25.3	19.2	12.0	8.6	5.0		
		MCWB	2.6	3.2	3.6	8.3	13.6	15.3	17.6	16.6	13.5	9.4	7.3	3.5		
	2%	DB	2.9	3.8	5.1	11.5	20.1	23.5	25.2	23.0	16.7	10.1	5.5	3.3		
		MCWB	1.6	2.2	2.5	5.9	11.7	14.1	16.2	15.7	12.5	8.6	4.5	2.1		
	5%	DB	1.0	1.9	3.8	9.0	17.1	20.7	23.2	20.7	14.7	8.7	3.8	1.8		
		MCWB	0.0	0.6	1.6	4.5	10.4	13.0	15.5	14.9	11.3	7.3	3.1	0.9		
	10%	DB	-0.5	0.1	2.4	6.8	14.0	18.5	21.2	18.5	13.2	7.4	2.4	0.7		
		MCWB	-1.1	-0.8	0.5	3.3	8.9	11.9	14.9	14.0	10.5	6.0	1.7	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	3.2	4.1	8.7	14.3	16.9	19.4	18.0	14.6	10.2	7.5	3.9		
		MCDB	4.1	4.8	6.6	15.3	20.9	22.9	24.1	22.6	17.8	11.3	8.5	4.8		
	2%	WB	1.6	2.3	2.8	6.4	12.7	15.1	17.6	16.6	13.2	8.8	4.7	2.2		
		MCDB	2.8	3.8	4.8	10.7	18.2	20.9	22.3	21.1	15.6	9.8	5.4	3.1		
	5%	WB	0.2	0.7	1.7	4.9	10.9	13.9	16.6	15.5	12.0	7.5	3.1	1.0		
		MCDB	0.9	1.7	3.6	8.3	16.6	19.1	21.5	19.5	14.1	8.6	3.6	1.6		
	10%	WB	-1.2	-0.8	0.7	3.7	9.1	12.7	15.6	14.4	11.0	6.2	1.7	0.1		
		MCDB	-0.6	0.0	2.3	6.5	13.5	17.3	20.0	18.1	12.7	7.2	2.2	0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.8	11.4	9.9	10.7	10.4	10.5	10.5	9.0	6.8	6.2	7.9		
		MCDBR	7.9	7.8	9.8	13.9	16.8	14.6	14.7	14.7	12.1	7.8	5.3	7.7		
		MCWBR	7.4	7.0	7.7	8.8	9.4	6.9	6.9	8.4	8.3	6.3	5.0	7.3		
		MCDBR	7.6	7.6	9.0	12.5	15.5	12.7	12.4	12.6	10.2	7.0	5.1	6.9		
	5% WB	MCWBR	7.1	6.9	7.3	8.2	9.2	6.5	6.6	7.5	7.5	5.9	4.8	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.188	0.260	0.289	0.323	0.330	0.329	0.341	0.329	0.303	0.285	0.337	N/A		
	taud		1.967	2.199	2.261	2.330	2.480	2.510	2.486	2.509	2.571	2.448	2.744	N/A		
	Ebn at Noon		369	654	797	852	883	890	865	844	800	639	355	N/A		
	Edh at Noon		24	55	82	97	92	92	92	81	62	45	21	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.13	0.70	2.09	3.81	4.71	5.25	4.80	3.60	2.06	0.85	0.20	0.05		
	RadStd		0.01	0.11	0.21	0.31	0.55	0.41	0.54	0.38	0.25	0.14	0.04	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page